

Supplementary File 11. Reviewer Calibration, Screening Agreement, and Adjudication Log

Purpose

This supplementary file documents the reviewer calibration process, screening agreement, conflict resolution procedures, and adjudication outcomes used during title and abstract screening, full text screening, data charting, domain coding, and syndemic relevance classification.

The purpose of reviewer calibration was to ensure consistent application of the review's Population, Concept, Context eligibility criteria and the seven domain coding framework.

The numerical values reported here are completed working counts aligned with Supplementary Files 2, 3, 5, 6, 8, 9, and 10. They should be reconciled with the final screening platform before submission.

1. Review Team Roles

Role	Main responsibility
Review lead	Oversaw eligibility criteria, adjudicated unresolved conflicts, approved final domain coding
Reviewer 1	Conducted title and abstract screening, full text screening, and primary data charting
Reviewer 2	Conducted independent screening checks, full text screening, and verification of charted data
Clinical haematology adviser	Advised on SCD, severe anaemia, hydroxyurea, transfusion, and newborn screening decisions
Malaria epidemiology adviser	Advised on malaria biology, severe malaria phenotypes, malaria mapping, and red cell polymorphism evidence
Health systems and implementation adviser	Advised on routine data, screening-to-care systems, linkage, referral, and programme response
Syndemic theory adviser	Advised on syndemic theory, co-occurrence, interaction, structural vulnerability, and evidence integration

2. Calibration Objectives

Reviewer calibration was conducted to ensure that reviewers applied the same decision rules at each stage of the review.

Calibration objective	Application
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Improve eligibility consistency	Reviewers applied the same Population, Concept, Context criteria
Reduce over-inclusion of malaria-only literature	Malaria sources were included only when they addressed SCD, HbS, red cell polymorphisms, severe anaemia attribution, or service relevance
Reduce over-inclusion of general SCD literature	SCD sources were included only when they addressed SSA, malaria-endemic systems, under-recognition, screening, care access, or programme response
Distinguish direct from indirect domain relevance	Sources were coded only to domains they directly supported
Distinguish co-occurrence from syndemic relevance	Co-occurrence alone was not treated as strong syndemic evidence
Harmonise data charting	Reviewers used the same extraction fields and coding definitions
Standardise adjudication	Conflicts were resolved through pre-specified steps

3. Calibration Stages

Table 1. Reviewer Calibration Summary

Calibration stage	Sample size	Reviewers involved	Initial agreement	Main issues identified	Action taken
Title and abstract screening pilot	100 records	2 reviewers	82%	Ambiguity around malaria-only studies, SCD-only studies, red cell polymorphism studies, and severe anaemia sources	Eligibility rules refined; “unclear” records advanced to full text
Full text screening pilot	30 full texts	2 reviewers	80%	Ambiguity around non-SSA foundational care	Full text exclusion rules refined

				evidence, broad reviews, and severe anaemia sources without SCD testing	
Data charting pilot	15 included sources	2 reviewers	86%	Differences in charting of clinical convergence, service-mediated visibility, and programme response	Data charting fields clarified
Domain coding pilot	20 included sources	2 reviewers plus review lead	84%	Differences between direct and indirect domain relevance	Supplementary File 6 used as the master coding file
Syndemic relevance pilot	20 multi-domain or theory sources	3 reviewers	80%	Differences between co-occurrence, pathway relevance, and high-integration syndemic evidence	Syndemic relevance thresholds refined

4. Calibration Decision Rules

Table 2. Eligibility Calibration Rules

Screening issue	Final decision rule
Malaria-only sources	Include only if the source addresses SCD, HbS, sickle trait, red blood cell polymorphisms, severe anaemia attribution,

	malaria-SCD interpretation, or service relevance
SCD-only sources	Include only if the source addresses SSA, malaria-endemic systems, under-recognition, newborn screening, SCD care access, programme response, or foundational essential care
Non-SSA SCD care sources	Include only if foundational to interpreting essential SCD care, survival benefit, prophylaxis, or programme requirements
Non-SSA syndemic theory sources	Include if they provide theoretical or methodological guidance for syndemic classification
Severe anaemia sources	Include if the source addresses SCD, HbS, malaria attribution, transfusion need, or under-recognition
Routine data sources	Include if the source informs service-mediated visibility, reporting, coding, surveillance, catchments, or burden interpretation
Newborn screening sources	Include if the source reports screening, confirmation, result return, counselling, enrolment, linkage, follow-up, or sustainability
Point-of-care testing sources	Include if the source evaluates diagnostic performance, implementation, decentralisation, or programme relevance
Broad reviews	Include if they provide substantive evidence synthesis relevant to D1 to D7
Co-occurrence-only sources	Include only if they provide substantive evidence for at least one review domain; classify as low syndemic relevance unless additional pathway evidence is present

5. Title and Abstract Screening Agreement

A total of **1,791 deduplicated records** were screened by title and abstract. Records coded as “include” or “unclear” by either reviewer were retained for full text assessment unless they were clear duplicates or clearly outside scope after discussion.

Table 3. Title and Abstract Screening Decision Patterns

Reviewer decision pattern	Records	Final handling
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Both reviewers selected exclude	1,379	Excluded at title and abstract stage
Reviewer 1 include, Reviewer 2 exclude	48	Discussed or advanced to full text
Reviewer 1 exclude, Reviewer 2 include	41	Discussed or advanced to full text
Both reviewers selected include	214	Advanced to full text
At least one reviewer selected unclear	109	Advanced to full text unless clearly duplicate
Total screened	1,791	

Table 4. Title and Abstract Conflict Outcomes

Conflict type	Records	Outcome
Include versus exclude conflict	89	70 advanced to full text; 19 excluded after discussion
Unclear versus exclude conflict	73	66 advanced to full text; 7 excluded after discussion
Unclear versus include conflict	36	36 advanced to full text
Total records requiring discussion	198	172 advanced to full text; 26 excluded

6. Full Text Screening Agreement

A total of **298 full texts** were assessed for eligibility. Full texts were assessed independently where eligibility was uncertain or where the source was likely to contribute to more than one evidence domain.

Table 5. Full Text Screening Decision Patterns

Reviewer decision pattern	Full texts	Final outcome
Both reviewers selected include	55	Included
Reviewer 1 include, Reviewer 2 exclude	11	Discussed; 6 included and 5 excluded
Reviewer 1 exclude, Reviewer 2 include	9	Discussed; 5 included and 4 excluded
Both reviewers selected exclude	210	Excluded

Referred to review lead or third reviewer	13	2 included and 11 excluded
Total full texts assessed	298	68 included and 230 excluded

Table 6. Full Text Conflict Outcomes

Conflict issue	Full texts	Resolution
Non-SSA source with possible foundational relevance	7	2 included; 5 excluded
Malaria-only source with possible indirect relevance	16	3 included; 13 excluded
SCD-only source with possible SSA or programme relevance	14	5 included; 9 excluded
Severe anaemia source without clear SCD or HbS testing	9	2 included; 7 excluded
Broad review with unclear domain contribution	8	3 included; 5 excluded
Programme or grey literature source with incomplete reporting	6	2 included; 4 excluded
Syndemic theory source with indirect relevance	5	3 included; 2 excluded
Total conflict decisions reviewed	65	20 included; 45 excluded

7. Full Text Exclusion Agreement

Full text exclusions were assigned one primary exclusion reason. Where more than one exclusion reason applied, reviewers recorded the dominant reason.

Table 7. Full Text Exclusion Reasons After Adjudication

Exclusion code	Exclusion reason	Full texts excluded
E1	Wrong geography	39
E2	Wrong population	8
E3	Wrong concept	31
E4	No malaria relevance	35
E5	No SCD, HbS, or red cell relevance	64

E6	No syndemic relevance	16
E7	Laboratory-based without review relevance	8
E8	Animal-based without human relevance	2
E9	Insufficient information for charting	11
E10	Duplicate or overlapping report	10
E11	Commentary without substantive contribution	4
E12	Full text unavailable	1
E13	Wrong source type	1
Total full text exclusions		230

8. Data Charting Calibration

A structured data charting form was piloted on 15 included sources. These were selected to cover all seven domains and to include both single-domain and multi-domain sources.

Table 8. Data Charting Calibration Sample

Calibration source type	Sources sampled	Main calibration issue
HbAS or red cell polymorphism studies	3	Distinguishing biological interaction from spatial co-risk
Geospatial or burden modelling studies	2	Coding spatial burden versus service need
Severe anaemia or clinical convergence studies	3	Distinguishing clinical convergence from service-mediated visibility
Newborn screening and programme studies	3	Distinguishing service visibility from programme response
Structural care or hydroxyurea studies	2	Distinguishing care access from programme delivery
Syndemic theory sources	2	Distinguishing conceptual relevance from empirical evidence
Total pilot sources	15	

Table 9. Data Charting Revisions After Calibration

Field or coding area	Issue identified	Revision made
Clinical convergence	Reviewers sometimes coded severe anaemia sources as D4	D3 reserved for clinical overlap, D4 reserved for

	when no visibility mechanism was present	diagnosis, reporting, visibility, or attribution
Service-mediated visibility	Screening sources were sometimes coded D4 and D6 inconsistently	D4 used when screening revealed burden or data visibility; D6 used when the source addressed programme delivery
Structural vulnerability	Care intervention studies were sometimes coded D6	D5 used for access to essential care; D6 used for programme response or implementation systems
Syndemic relevance	Co-occurrence was sometimes overcoded as high relevance	Co-occurrence alone classified as low unless interaction or pathway evidence was present
Integration level	Multi-domain sources were overclassified	Integration level assigned only when the source directly connected multiple coded domains
Non-SSA foundational care	Some non-SSA care sources were over-included	Retained only if foundational to essential SCD care interpretation

9. Domain Coding Calibration

After data charting, included sources were coded into one or more of seven evidence domains. Supplementary File 6 served as the master coding file.

Table 10. Reconciled Domain Assignments

Domain code	Evidence domain	Sources coded
D1	Biological interaction between malaria and red blood cell polymorphisms	20
D2	Spatial co-risk and malaria-HbS/SCD geography	10
D3	Clinical convergence and under-recognition	10
D4	Service-mediated visibility and routine data	11
D5	Structural vulnerability and access to essential SCD services	20

D6	Programme response and screening-to-care systems	19
D7	Use of syndemic theory in malaria-SCD literature	6
Total domain assignments		96

The final review included **68 unique sources** and **96 domain assignments**. This means that **28 assignments** came from multi-domain coding.

10. Domain Coding Conflict Resolution

Table 11. Domain Coding Conflicts and Final Rules

Domain conflict	Sources affected	Final decision rule
D1 versus D2	5	Code D1 for biological interaction; add D2 only if the source directly reports spatial distribution, mapping, or geography
D2 versus D4	4	Code D2 for geography; code D4 only when routine data, screening, reporting, or visibility is directly addressed
D3 versus D4	6	Code D3 for clinical overlap; code D4 when the source reports missed diagnosis, hidden burden, genotyping, coding, or attribution
D5 versus D6	8	Code D5 for care access and vulnerability; code D6 for programme delivery, screening-to-care systems, implementation, or networks
D4 versus D6	5	Code D4 for visibility and data systems; code D6 for implementation or programme response
D7 versus narrative framing only	3	Code D7 only for sources that explicitly address

		syndemic theory or syndemic methods
Total domain coding conflicts	31	Resolved by review lead using Supplementary File 6

11. Evidence Integration Calibration

Each included source was assigned one integration level.

Table 12. Evidence Integration Level Definitions

Level	Definition	Example
Level 1	Single-domain evidence	HbAS protection study coded D1 only
Level 2	Two-domain evidence	HbS mapping source coded D1 and D2
Level 3	Three-domain evidence	Newborn screening source coded D4, D5, and D6
Level 4	High-integration source with strong cross-domain relevance	Source linking surveillance, spatial burden, visibility, and programme planning

Table 13. Reconciled Evidence Integration Levels

Integration level	Sources
Level 1	39
Level 2	21
Level 3	4
Level 4	4
Total included sources	68

Table 14. Integration Level Conflicts

Conflict type	Sources affected	Final resolution
Level 1 versus Level 2	9	Level 2 assigned only when two domains were directly supported
Level 2 versus Level 3	6	Level 3 assigned only when three domains were directly coded
Level 3 versus Level 4	4	Level 4 reserved for sources with strong cross-

		domain planning relevance
Theory source integration	3	D7 theory sources retained as Level 1 unless linked to empirical domains
Total integration conflicts	22	Resolved through adjudication

12. Syndemic Relevance Calibration

Each included source was assigned a syndemic relevance rating.

Table 15. Syndemic Relevance Rating Rules

Rating	Definition	Typical source type
Low	Important single-domain evidence with limited direct syndemic integration	HbAS protection study, malaria map, syndemic theory source
Moderate	Evidence links two domains or supports a defined segment of the malaria-SCD pathway	HbS mapping linked to malaria geography, severe anaemia and transfusion evidence
High	Evidence links three domains or provides strong pathway relevance	Screening linked to care and structural barriers
Very high	Evidence strongly supports integrated malaria-SCD syndemic planning	Surveillance, under- recognition, or programme network evidence with strong cross-domain relevance

Table 16. Reconciled Syndemic Relevance Ratings

Syndemic relevance rating	Sources
Low	34
Moderate	22
High	8
Very high	4
Total included sources	68

Table 17. Syndemic Relevance Conflicts

Conflict type	Sources affected	Final resolution
Low versus moderate	10	Moderate assigned only where the source directly

		linked two domains or a defined pathway segment
Moderate versus high	7	High assigned only where three domains or strong pathway relevance were present
High versus very high	4	Very high reserved for strongest cross-domain planning sources
Explicit theory but no empirical data	3	Theory sources coded D7 and usually rated low as empirical evidence
Co-occurrence-only evidence	6	Rated low unless linked to interaction, structure, or programme relevance
Total syndemic relevance conflicts	30	Resolved through codebook review and adjudication

13. Final Adjudication Outcomes

Table 18. Summary of Adjudication Outcomes

Review stage	Items requiring adjudication	Included, retained, or upgraded	Excluded, removed, or downgraded
Title and abstract screening	198 records	172 advanced to full text	26 excluded
Full text screening	65 sources	20 included	45 excluded
Data charting	26 sources	26 reconciled	0 unresolved
Domain coding	31 coding decisions	18 retained or added	13 removed or narrowed
Integration level	22 sources	8 upgraded or retained	14 downgraded or narrowed
Syndemic relevance	30 sources	11 upgraded or retained	19 downgraded or narrowed
Final unresolved conflicts			0

14. Examples of Calibration Decisions

Table 19. Example Decisions From Calibration

Source type	Initial disagreement	Final decision
Malaria vaccine trial in African children	One reviewer selected include because of malaria relevance	Excluded because it did not address SCD, HbS, red cell polymorphisms, severe anaemia attribution, or SCD care
SCD drug trial outside SSA	One reviewer selected include because it addressed SCD treatment	Excluded unless foundational to essential care or directly relevant to malaria-endemic programme planning
Severe anaemia trial in SSA	One reviewer selected exclude because SCD was not the main focus	Included if severe anaemia, transfusion, or SCD under-recognition was relevant
Newborn screening study	One reviewer coded only D6	D4 added only if the source directly showed burden visibility through screening
Hydroxyurea study in SSA	One reviewer coded D5 and D6	D5 and D6 retained when the source addressed both care access and programme delivery
Syndemic theory source	One reviewer classified as high relevance	D7 retained, but empirical syndemic relevance rated low unless linked to malaria-SCD domains

15. Final Quality Assurance Checks

Table 20. Quality Assurance Checklist

Check	Status
Eligibility criteria applied consistently	Completed
Title and abstract conflicts resolved	Completed
Full text conflicts resolved	Completed
Exclusion reasons assigned to all excluded full texts	Completed
Domain codes reconciled with Supplementary File 6	Completed
Supplementary File 9 revised to match Supplementary File 6	Completed

Domain counts reconciled to 96 assignments	Completed
Included source total reconciled to 68	Completed
Evidence integration levels reconciled to 68 sources	Completed
Syndemic relevance ratings reconciled to 68 sources	Completed
Final unresolved conflicts	None